

Composition of blood And Cellular elements

LEARNING OBJECTIVES

- Composition of blood
- Functions of blood.
- Plasma and it's composition.
- R.B.Cs: Their structure, function and count.
- W.B.Cs: Their types, structure, function and count.
- Platelets: Their structure, function and count.

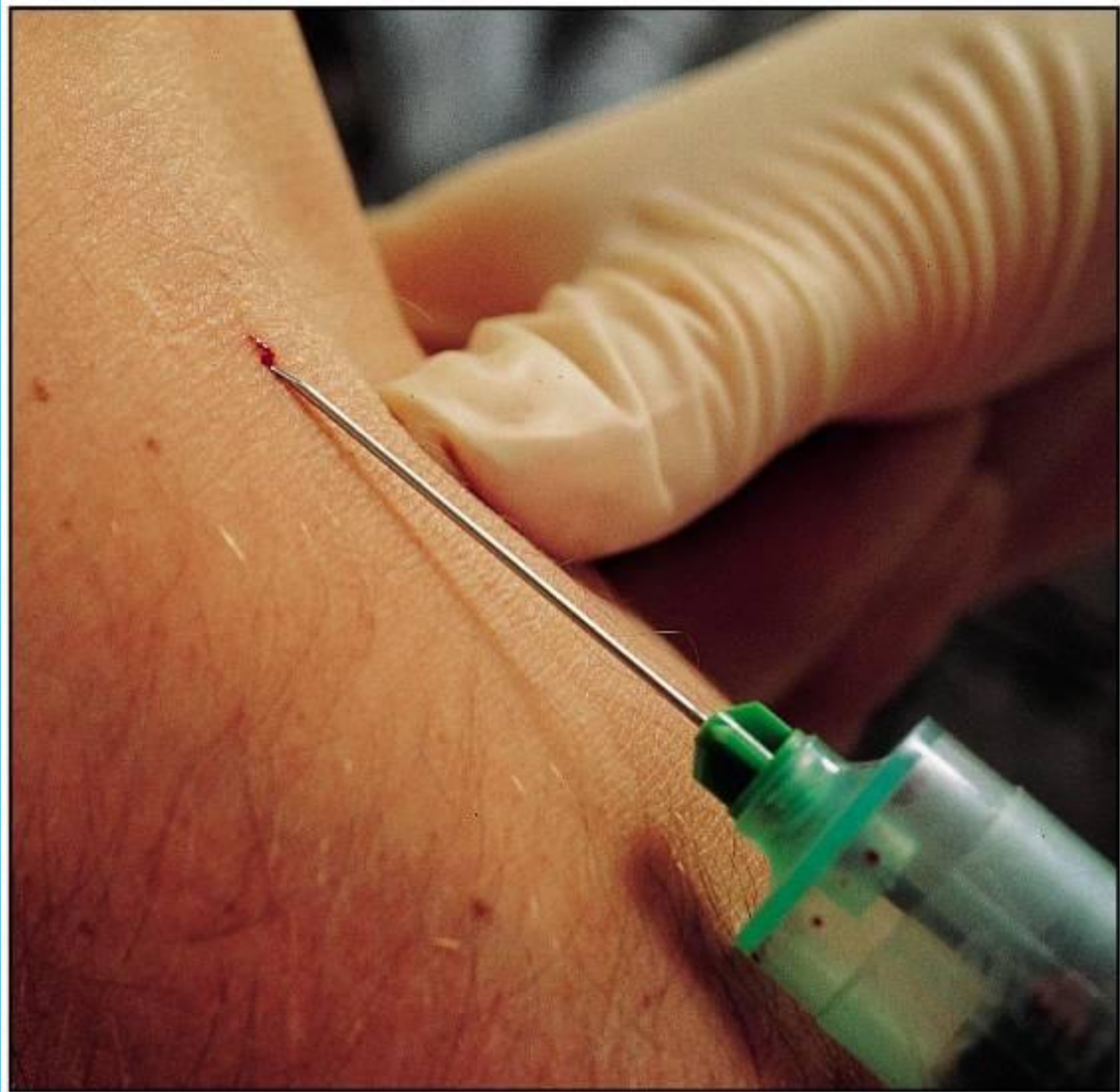
Blood

- Composition
 - Formed elements:
 - Cells
 - Plasma:
 - Fluid

The Functions of Blood

- Blood functions include:
 - Transport of cells and compounds.
 - Regulate pH and electrolytes of *interstitial fluids*.
 - Limit blood loss through damaged vessels.
 - Defend against pathogens and toxins.
 - Absorb and distribute heat as a part of temperature regulation.

•The Composition of Whole Blood

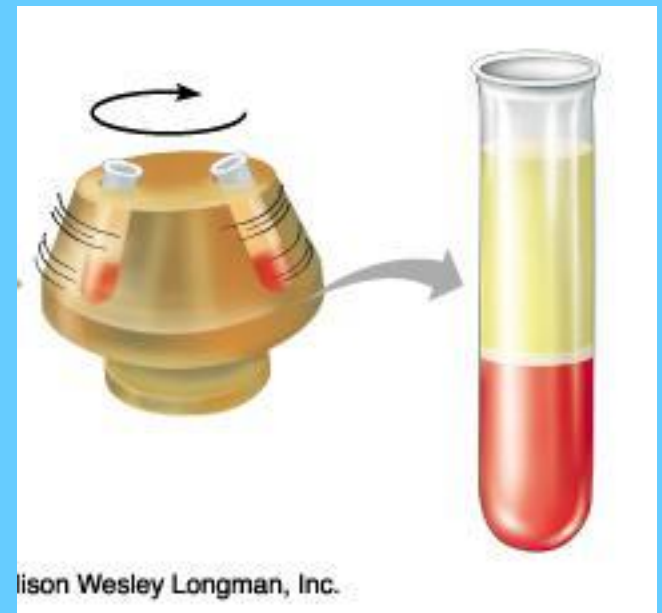
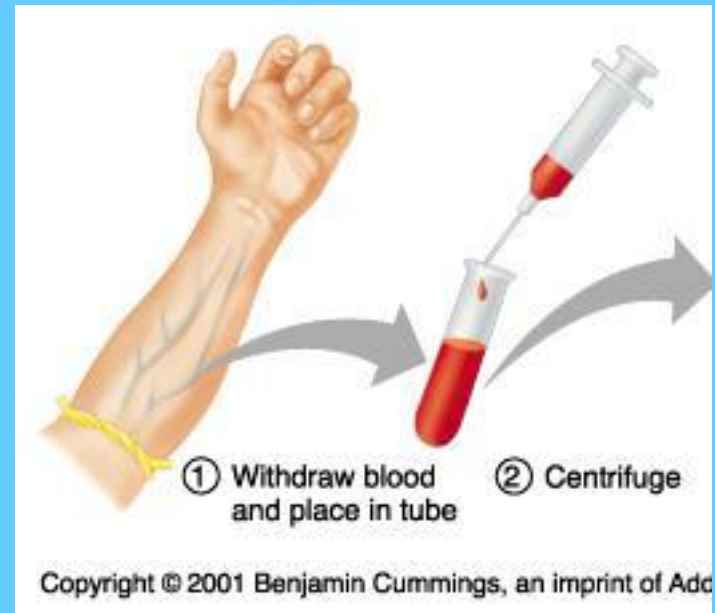


(a) Taking a blood sample

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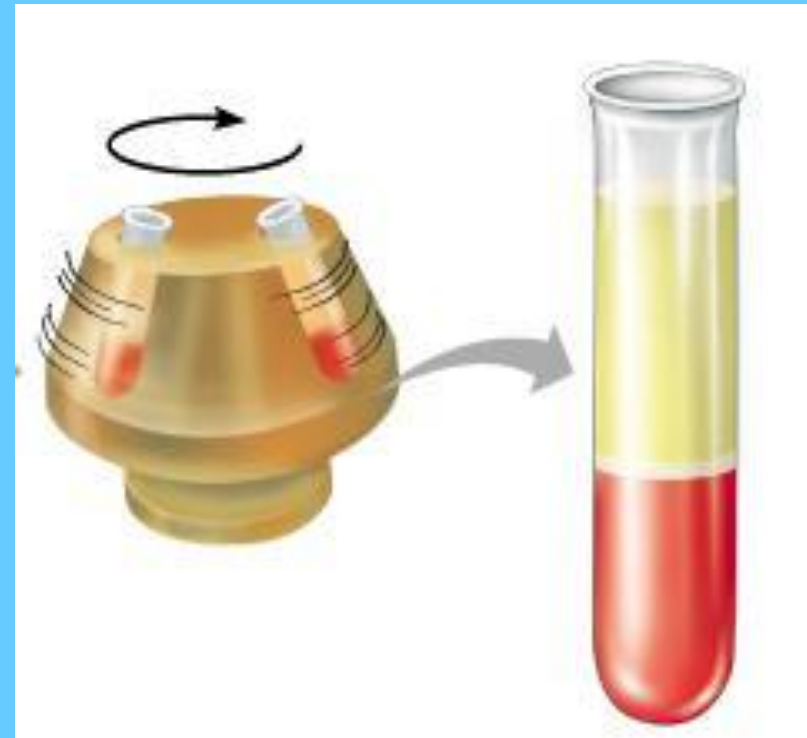
Separated blood

- Centrifuged blood:
 - Plasma 55%
 - Buffy coat
 - WBCs <1%
 - Erythrocytes
 - Hematocrit
 - 45%



Plasma

- 90% water
- 10% solutes
 - Ions, e.g., Na^+ , Cl^- , Ca^{++}
 - Nutrients, e.g., simple sugars, amino acids, lipids
 - Wastes, e.g., urea, ammonia, CO_2
 - Miscellaneous: O_2 , hormones, vitamins, plasma proteins



Plasma proteins

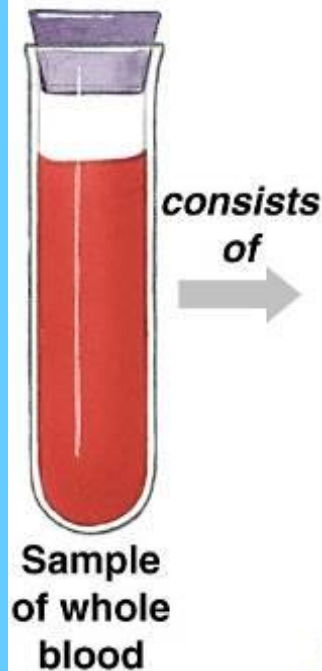
- Albumin
 - Helps control osmotic pressure
 - Helps control diffusion of water (recall edema)
- Globulin
 - Includes antibodies (Abs)
 - Transport proteins (lipids, iron, copper, etc.)
- Fibrinogen
 - Involved in clotting

Serum

- Plasma with clotting factors removed
- Let blood sit, pour off supernatant

Plasma

• The Composition of Whole Blood



consists of

PLASMA COMPOSITION	
Plasma proteins	7%
Other solutes	1%
Water	92%
Transports organic and inorganic molecules, formed elements, and heat	

(b) Components of plasma

Plasma (55%)

PLASMA PROTEINS

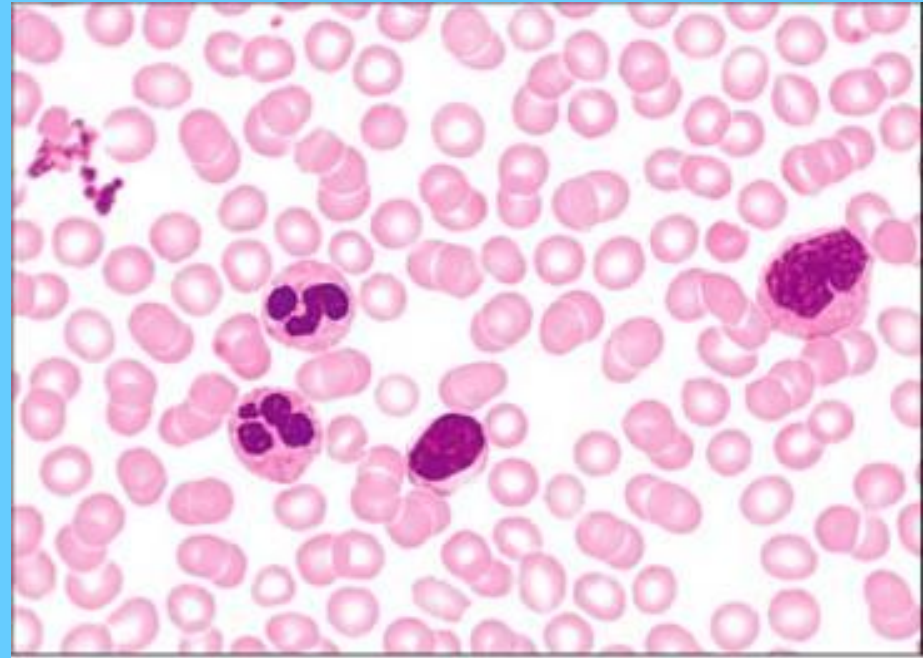
Albumins (60%)	Major contributors to osmotic pressure of plasma; transport lipids, steroid hormones
Globulins (35%)	Transport ions, hormones, lipids; immune function
Fibrinogen (4%)	Essential component of clotting system; can be converted to insoluble fibrin
Regulatory proteins (<1%)	Enzymes, proenzymes, hormones

OTHER SOLUTES

Electrolytes	Normal extracellular fluid ion composition essential for vital cellular activities. Examples: Na^+ , K^+ , Ca^{2+} , Cl^- , HCO_3^-
Organic nutrients	Used for ATP production, growth, and maintenance of cells. Examples: Fatty acids, glucose, amino acids
Organic wastes	Carried to sites of breakdown or excretion. Examples: Urea, bilirubin

Formed elements

- RBCs (erythrocytes)
- WBCs (leukocytes)
 - Granulocytes
 - Agranulocytes
- Blood smear
 - Light microscope
 - Stained



Erythrocytes

- 7-8 μm diameter
- Biconcave disc shape
 - $\uparrow$ surface area
 - $\uparrow$ efficiency for diffusion of O_2 & CO_2



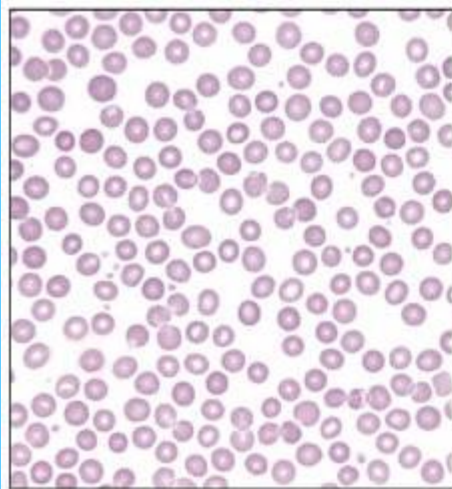
Cross-sectional view



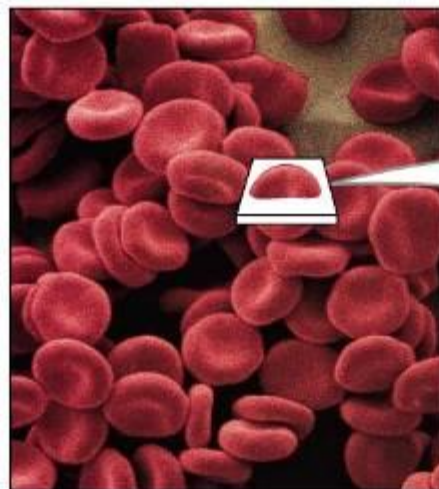
Top view

Formed Elements

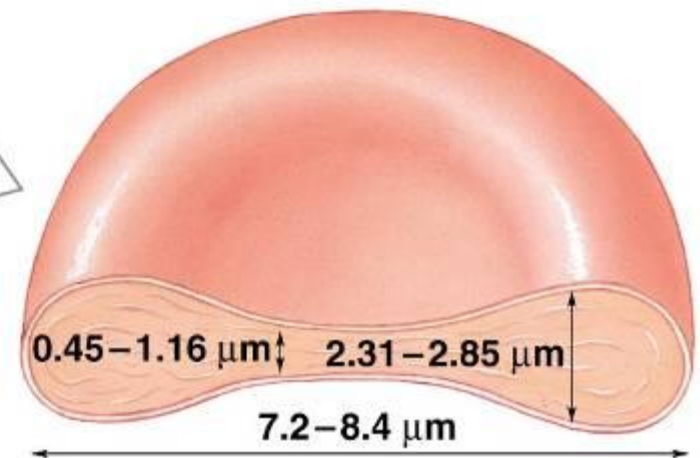
- The Anatomy of Red Blood Cells



(a)



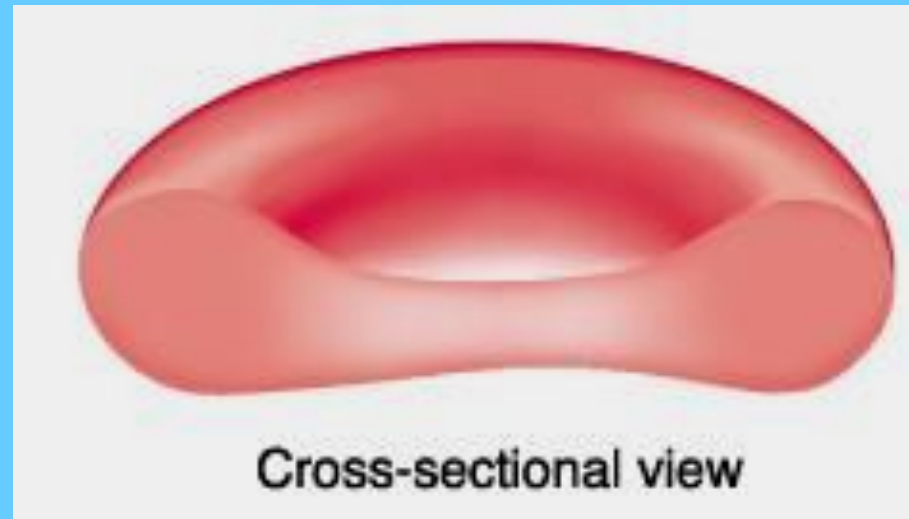
(b)



(c)

Erythrocytes

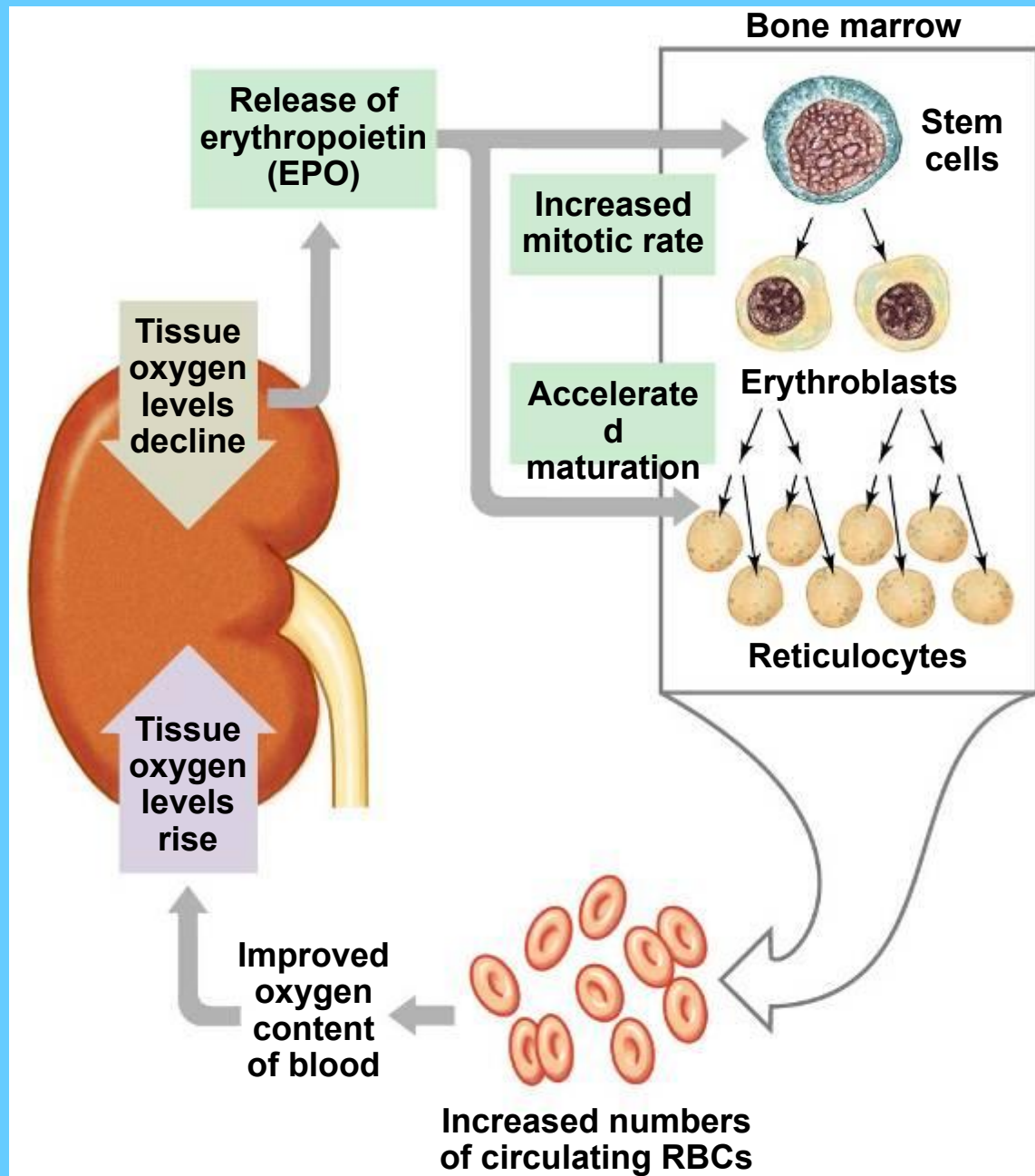
- Structure
 - Plasma membrane
 - Cytoplasm
 - Hemoglobin
 - Binds O_2 & CO_2
 - No nucleus or organelles



Erythrocytes

- Flexible
- Elastic
- 100-120 day life span
- Originate in bone marrow





Formed Elements

- Key Note**

- Red blood cells (RBCs) are the most numerous cells in the body. They circulate for about four months before being recycled; millions are produced each *second*. The hemoglobin inside transports oxygen from the lungs to peripheral tissues and carbon dioxide from the tissues to the lungs.

Formed Elements

Properties of RBCs

- Transport oxygen and carbon dioxide in blood stream

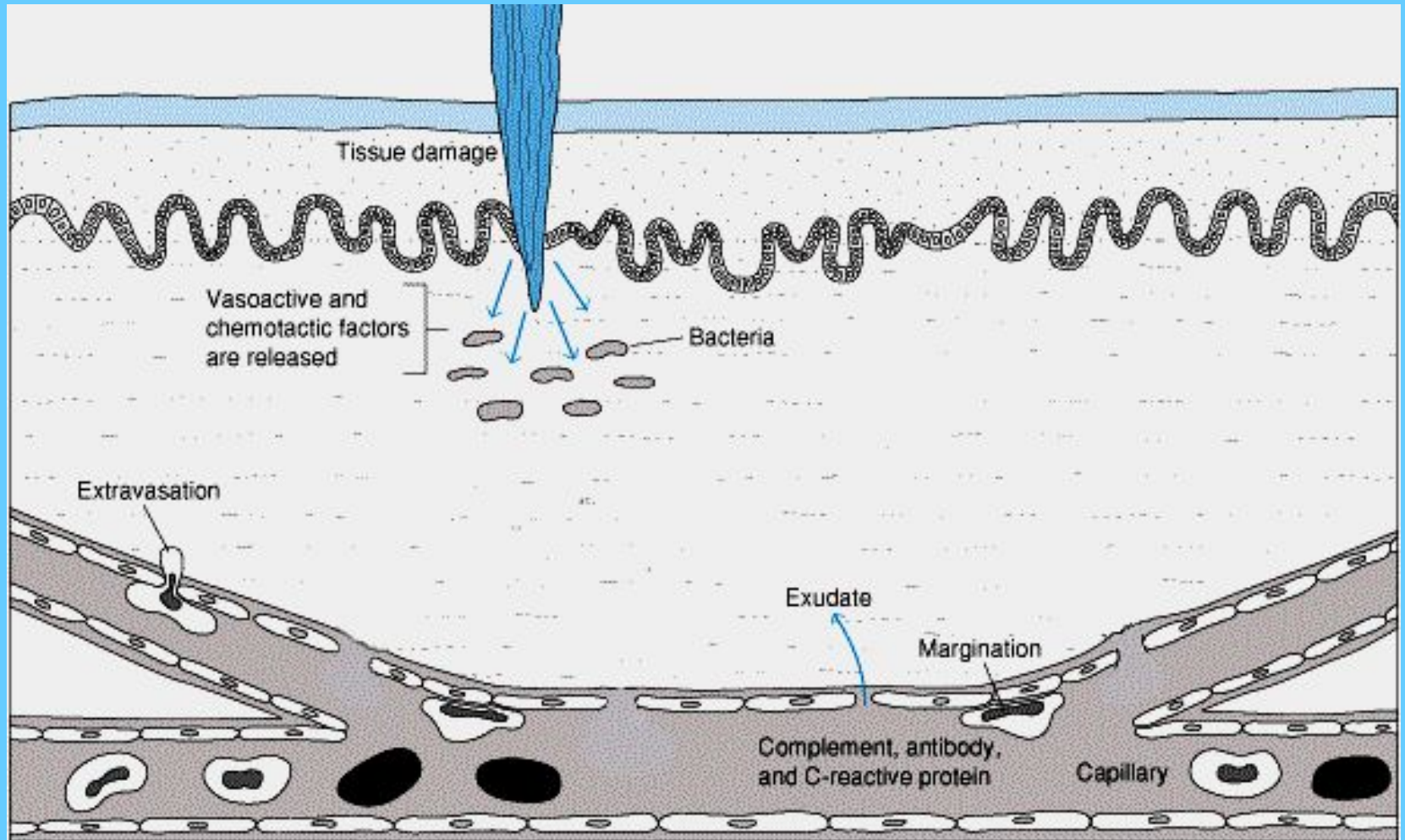
- Have large surface to volume ratio
 - Speeds up gas loading/unloading

- Lack most organelles
 - Makes more room for *hemoglobin*
- Degenerate after about 120 days

Leukocytes (WBCs)

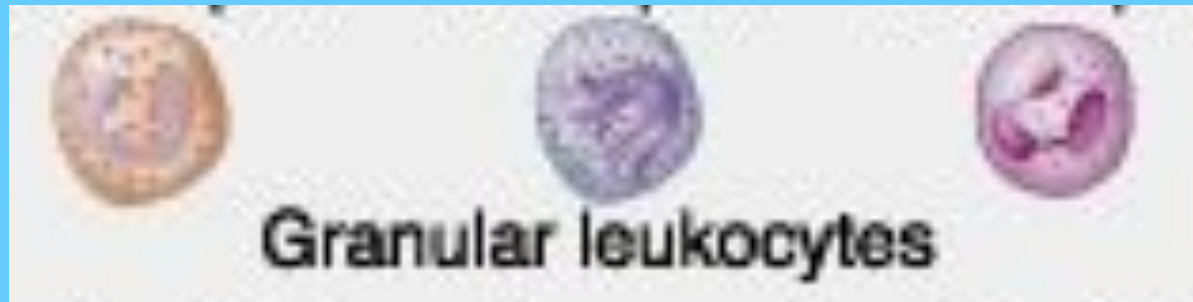
- Part of defense system
 - Protect against bacteria, viruses, parasites
- Attracted to sites of infection
- Diapedesis: leave capillary by squeezing between endothelial cells
- Ameboid movement
 - Travel toward infection
- Exhibit *chemotaxis*—Move toward specific chemicals released by bacteria or injured cells
- Originate in bone marrow
- Granulocytes / agranulocytes

Diapedesis



Granulocytes

- WBCs with granules in cytoplasm
 - Visible with LM
- Neutrophils
- Eosinophils
- Basophils
- Phagocytic
- Larger than RBCs
- Lobed nuclei



Neutrophils

- 60% of WBCs
- Lobed nucleus
- Light staining granules
 - Digestive enzymes
- Function
 - Phagocytize & destroy bacteria
- First cells to respond to infection
- Secrete antibacterial chemicals
- Phagocytize & digest bacteria



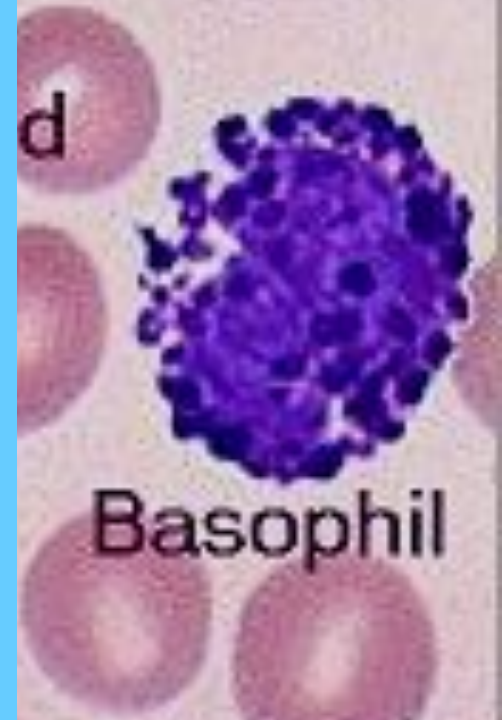
Eosinophils

- 1-4% of WBCs
- Lobed nucleus
- Eosin-staining granules
- Phagocytize allergen-Ab complexes
- Secrete antihistamine
- Attack parasites



Basophils

- 0.5% of WBCs
- Lobed nucleus
- Large granules stained dark purple
- Granules
 - Histamine – creates inflammation in allergic reaction



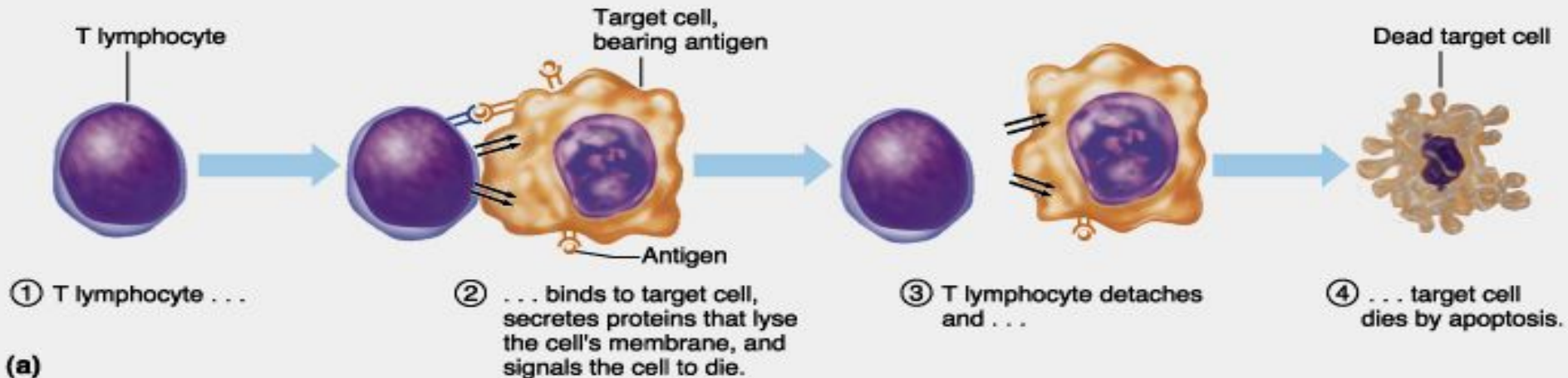
Lymphocytes

- Agranulocyte
- 20-45% of WBCs
- Spherical, dark-staining nucleus
- Thin rim of blue staining cytoplasm
- Each lymphocyte recognizes and acts against a specific antigen



Lymphocytes

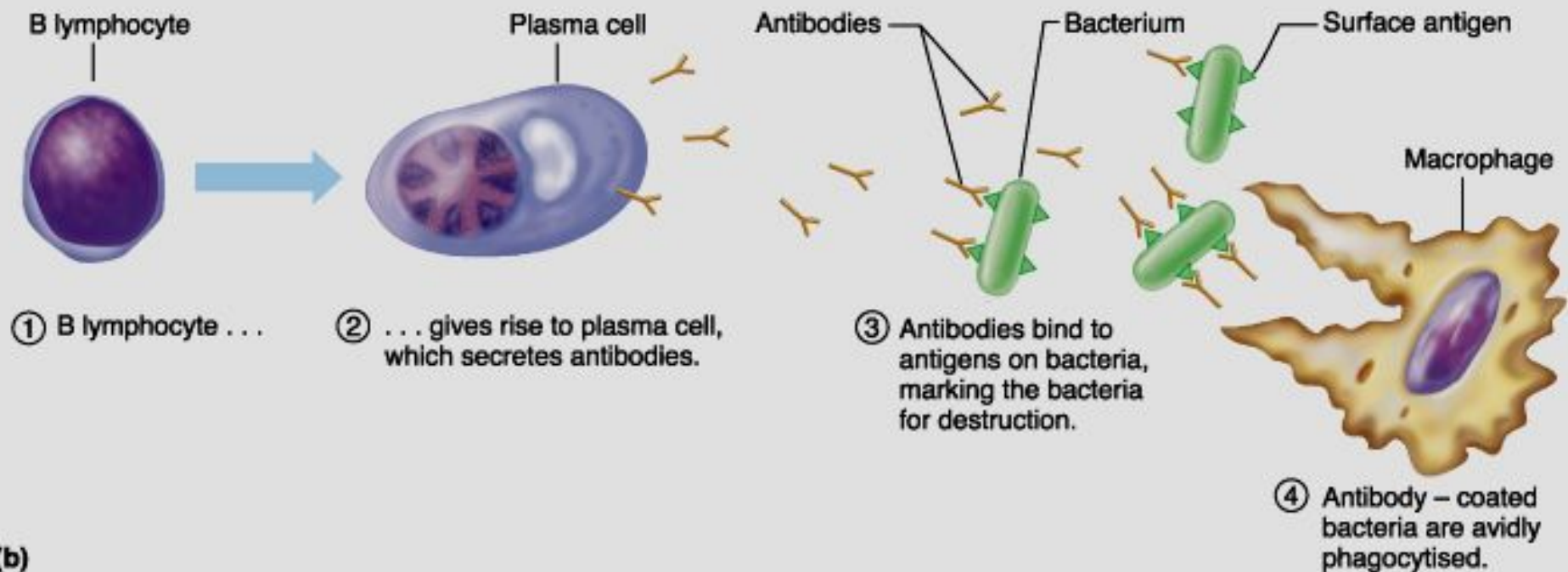
- T lymphocytes can attack foreign cells directly



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Lymphocytes

- B lymphocytes transform into plasma cells and secrete antibodies



(b)

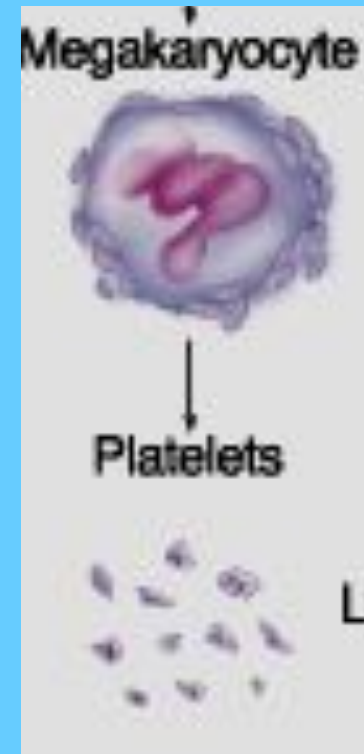
Monocytes

- Agranulocyte
- 4-8% of WBCs
- Horseshoe shaped nucleus
- Grey-blue stained cytoplasm
- Become wandering macrophages after diapedesis



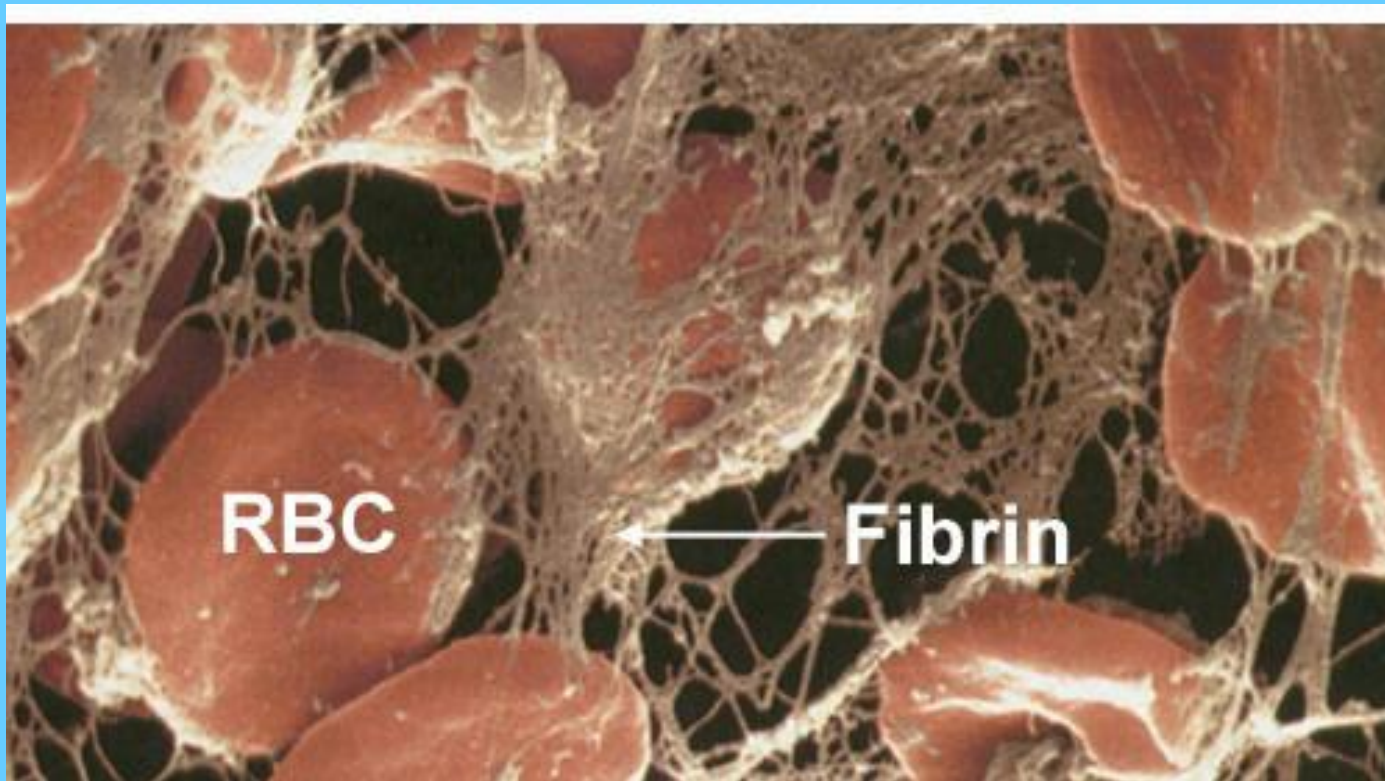
Platelets

- (thrombocytes)
- Fragments of megakaryocytes in bone marrow
- Attracted to hemorrhage
 - Plugs leaks
 - Promotes constriction of blood vessel
 - Triggers inflammation
 - Initiates clotting reaction



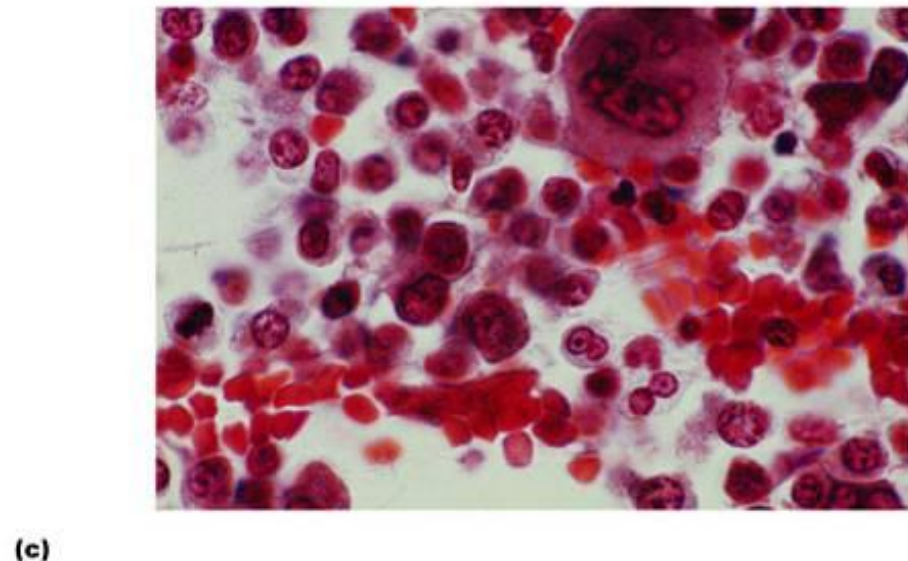
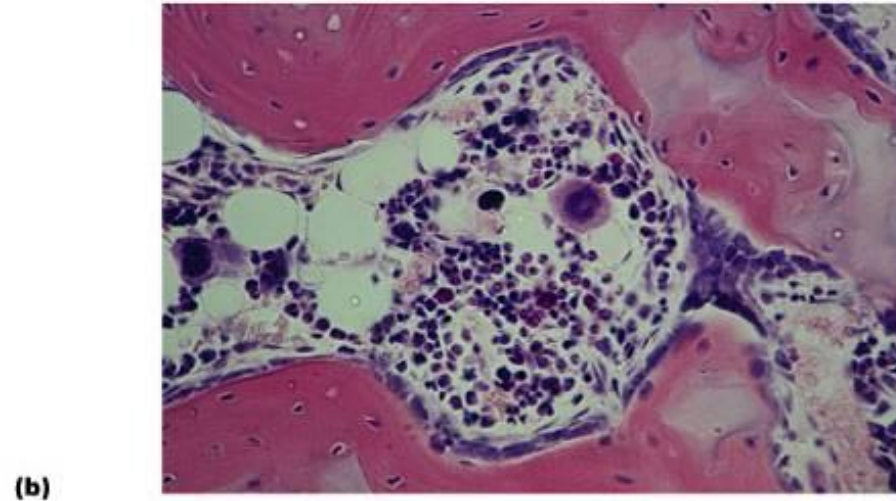
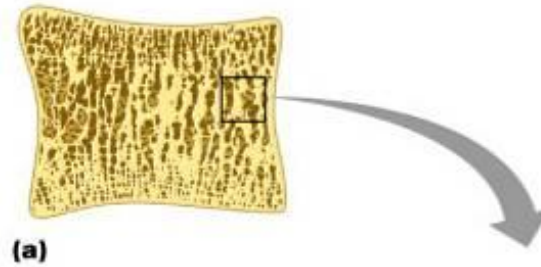
Platelets

- SEM of a clot with platelet, fibrin mesh, rbc's



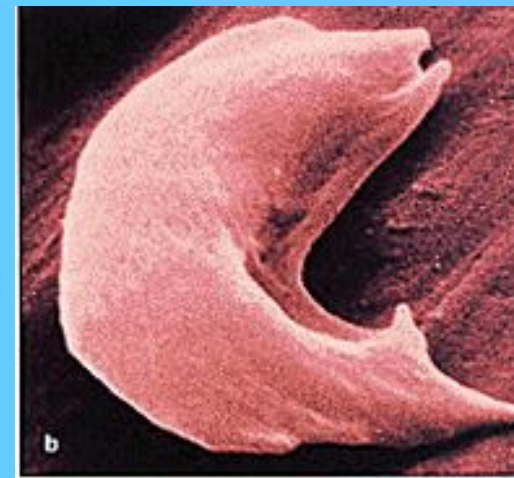
Hematopoiesis

- Occurs in red marrow
- Red marrow replaced by yellow in limbs between 8-18 yrs
- Blood stem cells
 - Pluripotent
- Myeloid stem cell
- Lymphoid stem cells



Sickle cell disease

- Genetic condition
- 1 nucleotide substitution
- 1 amino acid substitution
- Hbs
- With low O_2 Hbs polymerizes
- Creates “sickle” shape



Sickle cell disease

- ↓ flexibility
- ↑ fragility
- ↑ blood viscosity
- ↓ O_2
- ↑ sickling
- “crisis”
- Painful ischemia
 - Lack of O_2



Disorders of the Blood

- Disorders of leukocytes
 - Leukemia – a form of cancer
 - Classified as **lymphoblastic** or **myeloblastic**
- Disorders of platelets
 - Thrombocytopenia
 - Abnormally low concentration of platelets

The Blood Throughout Life

- First blood cells develop with the earliest blood vessels
- Mesenchyme cells cluster into blood islands
- Late in the second month
 - Liver and spleen take over blood formation
- Bone marrow becomes major hematopoietic organ at month 7

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